Lab Manual: Information Security

Lab Title: Kali Linux Setup: Configure & test a vulnerable Windows 7 target

Lab Objectives

By the end of this lab students will be able to:

7. Mount and run a Kali Linux VM image in VirtualBox (OVA/ISO) without performing a full installation.
8. Create and configure a Windows 7 VM to act as a vulnerable target.
9. Configure an isolated virtual network (Host-only or Internal) so the Kali and Win7 VMs can communicate.
10. Use basic terminal/command-line tools to discover VM IP addresses and verify connectivity.
11. Understand and follow lab safety/ethics for working with intentionally vulnerable images.

Learning Outcomes (CLOs & PLOs Mapping)

CLO	CLO Statement	Bloom's Level	PLO
1	Apply Information Security techniques into practice.	P3 (Apply)	PLO-5

Lab Overview

This lab demonstrates how to set up a small, isolated penetration-testing environment using VirtualBox. You will mount a prebuilt Kali image (OVA/ISO) in VirtualBox and create a Windows 7 VM to serve as a vulnerable target. Both VMs will be attached to an isolated virtual network so students can practice reconnaissance and connectivity checks without risk to other networks.

The lab focuses only on installation/mounting, networking, and simple verification (IP and ping). **No active exploitation will be performed in this session.**

Intended complexity: beginner-friendly, installation + basic networking checks only.

What is already done for you
The **Kali VM is already set up** and you already know how to mount an ISO in VirtualBox, so those steps are not repeated here. In this lab you only (1) check Kali's network setting, (2) create the Windows 7 target, and (3) test that the two machines can see each other.

Lab Safety and Ethics

Windows 7 is an old, intentionally vulnerable system. Treat it like a sample in a chemistry lab: useful for learning, dangerous if it escapes.

- **Keep the target isolated.** Use only a Host-only or Internal network. Never attach the Windows 7 VM to NAT or Bridged mode, and never connect it to the Internet.
- **Scan only your own lab.** Run discovery and port scans only against the two VMs in this lab. Scanning the university network, home Wi-Fi or any other device without written permission is unethical and can be illegal.
- **No exploitation today.** This session is limited to installation, networking, IP discovery and ping.
- **Take a snapshot** of the Windows 7 VM after setup (VirtualBox → Machine → Take Snapshot) so you can return to a clean state.
- **Do not reuse real passwords** inside lab VMs, and do not copy files from the vulnerable VM to your real computer.

Key Concepts Explained Simply

Read this section first. Every command in the lab uses one of these ideas.

1. Virtual Machine, Host and Guest

A **virtual machine (VM)** is a computer simulated by software. **VirtualBox** is the program that creates it. The real computer you sit at is the **host**; each VM running inside VirtualBox is a **guest**. In this lab you have two guests: Kali (the attacker's machine) and Windows 7 (the target).

2. VirtualBox Network Modes

Each VM has a virtual network card (an *adapter*). The mode you pick decides who it can talk to:

Mode	VM ↔ other VM	VM ↔ host	VM → Internet	Use in this lab?
NAT	No	No (host cannot start contact)	Yes	No
Bridged	Yes (on your real network)	Yes	Yes	Never
Host-only	Yes	Yes	No	Yes (recommended)
Internal Network	Yes (same network name)	No	No	Yes (alternative)

Why isolation? Host-only and Internal networks have no route to the Internet, so the vulnerable Windows 7 machine cannot be reached from outside and cannot attack anyone.

3. IP Address, Subnet and Gateway

An **IP address** (like 192.168.56.101) is the address of a machine on a network, like a house number. Machines on the same street share the first part. The **subnet** says which part is the street. **192.168.56.0/24** (mask 255.255.255.0) means the first three numbers (192.168.56) are the street and the last number (1 to 254) identifies the machine. A **gateway** is the door out of the street to other networks (on a host-only network there may be none, and that is fine).

4. DHCP and Static IP

DHCP automatically hands out IP addresses when a machine starts. **Static** means you type the address yourself.

- **Host-only network:** VirtualBox runs a DHCP server by default, so the VMs normally get addresses like 192.168.56.101 and .102 automatically.
- **Internal network:** there is **no DHCP server**. Each VM will have no IP until you set one by hand (see Troubleshooting).

5. MAC Address and ARP

Every network card has a unique hardware label called a **MAC address** (like 42:9d:cc:96:de:ea). On a local network, machines actually deliver data using MAC addresses. **ARP** (Address Resolution Protocol) is the phone book that links IP addresses to MAC addresses. A machine fills its ARP table only after it has talked to another machine.

6. Ping and ICMP

Ping sends a small “are you there?” message (ICMP echo request) and waits for an echo reply. If replies come back, the two machines can reach each other. Windows 7's firewall blocks ping by default, which is why a ping to Windows 7 may fail even when the network is correct.

7. Ports and Services

If an IP address is a building, **ports** are its doors. Each service listens at its own door. Examples: **80** = web (HTTP), **445** = Windows file sharing (SMB), **3389** = Remote Desktop (RDP). A port is **open** if a service is listening, **closed** if nothing is listening and **filtered** if a firewall hides it.

8. Reconnaissance

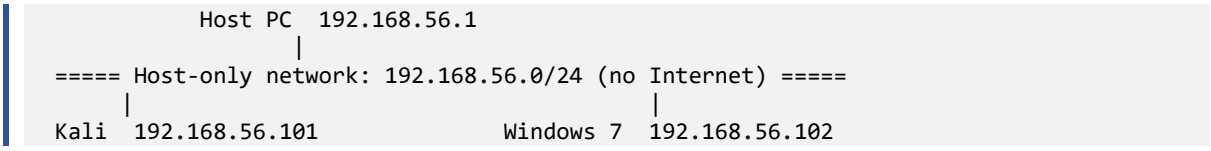
Before testing a system, you first find out what exists: which machines are alive (host discovery) and which ports are open (port scan). This is called **reconnaissance**. Today you only do this safely inside your own lab.

Quick Network Design (Recommended)

Use a Host-only or Internal network (name it **lab-net** / **vbox-lab**) to isolate lab traffic from your host/internet.

Example addressing (Host-only DHCP, or set static):

Machine	Role	Example IP
Kali (DHCP/static)	Attacker / tester	192.168.56.101
Windows 7 (DHCP/static)	Vulnerable target	192.168.56.102
Host (VirtualBox host-only adapter)	Your real computer	192.168.56.1



Important

Your actual IP addresses may be different. DHCP gives the first free numbers (often .101 and .102, but not guaranteed). **Always check your own IP with the commands below and use your own addresses.**

Step-by-Step Manual

Part A: Kali VM (Already Set Up): Quick Check Only

Do not reinstall or re-mount anything. Just confirm the following in VirtualBox (select Kali → Settings):

Where	Setting	Why
System → Processor	2 or more CPUs	Kali's tools run faster and the desktop stays smooth.
System → Motherboard	4 GB or more RAM (if your PC allows)	Kali is a full desktop system and needs memory.
Network → Adapter 1	Attached to: Host-only Adapter (vboxnet0) or Internal Network (lab-net)	Puts Kali on the isolated lab network.

Do not start Kali yet; first create the Windows 7 target in Part B.

Part B: Create the Windows 7 VM (Vulnerable Target)

1. **Create the VM.** In VirtualBox click **New**. Give it a name such as Win7-Target, choose Type **Microsoft Windows** and Version **Windows 7 (32-bit or 64-bit)**. The version must match your image (x86 = 32-bit, x64 = 64-bit).
2. **Give it resources.** Suggested: 2 GB RAM (2048 MB) and a 25 GB virtual disk. Windows 7 is old and runs well with modest resources.
3. **Attach the image.** Attach the Windows 7 ISO (Settings → Storage → empty optical drive) **or** import the OVA (File → Import Appliance). You already know how to do this from the Kali setup.
4. **Set the network.** Settings → Network → Adapter 1 → Attached to the **same** Host-only Adapter / Internal Network used for Kali. Leave Adapters 2 to 4 disabled. This is what lets the two VMs talk and keeps the target away from the Internet.
5. **Install or boot Windows 7.** Start the VM and complete only the minimal setup (a user name and password, no online updates).
6. **Take a snapshot** named “clean-setup”.

Remember
Do **not** connect the target VM to the Internet. Keep it isolated.

Part C: Verify VirtualBox Network Settings

Do this check for **both** VMs:

1. Select the VM → **Settings** → **Network** → **Adapter 1**.
2. Check “Attached to” is **Host-only Adapter** (or **Internal Network**).
3. Check both VMs use the **same** host-only/internal network name. If the names differ, the VMs are on different “streets” and can never reach each other.

4. For Host-only, make sure the adapter exists on the host: File → Host Network Manager (in VirtualBox 7: File → Tools → Network Manager). You should see vboxnet0 with address 192.168.56.1 and DHCP enabled.

Part D: Start Both VMs and Check Basic Networking

Start Kali and Windows 7. Wait until both desktops appear. First run the commands on Kali, then on Windows 7.

> **About the sample outputs**
> The sample outputs in this manual were produced by running these exact commands on a test lab with the same addressing (Kali 192.168.56.101, Windows 7 192.168.56.102, host 192.168.56.1). Your MAC addresses, timings and some details will differ.

D1. On Kali (Terminal)

Open a terminal and run the commands in this order.

1. ip a

In one line: Lists every network card and its IP address.

Command

```
ip a
```

What to look for

- **lo** is the loopback card (127.0.0.1) and always exists.
- **eth0** is the VirtualBox network card (the name may differ; use the name you see).
- **inet 192.168.56.101/24** is your IPv4 address; /24 is the subnet.
- **link/ether** is your MAC address. **state UP** means the card is working.

Sample output (tested)

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 ... state UNKNOWN
    inet 127.0.0.1/8 scope host lo
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 ... state UP
    link/ether 42:9d:cc:96:de:ea brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 scope global eth0
```

2. hostname -I

In one line: Prints only the IP address(es) of this machine.

Command

```
hostname -I
```

What to look for

- A short way to get your IP without the extra details.

Sample output (tested)

```
192.168.56.101
```

3. ip route

In one line: Shows the routing table: which network card is used to reach which network.

Command

```
ip route
```

What to look for

- The line **192.168.56.0/24 dev eth0** means the lab network is reached through eth0.
- A line starting with **default via** is the gateway. On a real Host-only network you may see **only** the 192.168.56.0/24 line. That is normal because there is no Internet.

Sample output (tested)

```
default via 192.168.56.1 dev eth0
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.101
```

4. arp -n

In one line: Shows the ARP table: the IP-to-MAC pairs this machine has learned (-n = show numbers, do not look up names).

Command

```
arp -n
```

What to look for

- Right after boot the table may be **empty**. Run it again after the ping in the next step and you will see Windows 7's entry. This proves the two machines have talked.

Sample output (tested)

```
(before ping)  (no entries)

(after ping)
Address         HWtype  HWaddress           Flags Mask  Iface
192.168.56.102  ether   26:b5:08:6c:1d:ab   C           eth0
```

5. ping -c 4 192.168.56.102

In one line: Sends 4 test messages to Windows 7 and reports whether it replies (-c 4 = stop after 4).

Command

```
ping -c 4 192.168.56.102
```

What to look for

- Replace 192.168.56.102 with **your** Windows 7 IP.
- **4 received, 0% packet loss** means connectivity works.
- Linux machines reply with TTL=64; a real Windows machine usually replies with **TTL=128**.
- You can also ping the host: ping -c 4 192.168.56.1

Sample output (tested)

```
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.054 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.252 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.063 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.063 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
```

6. Ping Sweep (Find Live Hosts)

In one line: Pings every address from .1 to .254 one by one and prints the ones that answer.

Command

```
for ip in 192.168.56.{1..254}; do ping -c1 -W1 $ip &>/dev/null && echo "up: $ip"; done
```

How to read this command

Part	Meaning
for ip in 192.168.56.{1..254}	Repeat for every address 192.168.56.1, .2 ... .254 (stored in ip).
ping -c1 -W1 $ip	Send one ping (-c1) and wait at most 1 second for a reply (-W1).
&>/dev/null	Hide all ping output (send it to the “bin”).
&& echo "up: $ip"	Only if the ping succeeded (&&), print that address as up.

Tested on a small range, the output was:

```
up: 192.168.56.101
up: 192.168.56.102
```

Be patient, or use the faster version

The loop pings one address at a time, and every empty address wastes 1 second, so the full sweep takes about **4 minutes**. A faster version runs all pings together (tested: about 1.5 seconds):

```
for ip in 192.168.56.{1..254}; do (ping -c1 -W1 $ip &>/dev/null && echo "up: $ip") &
done; wait
```

The **&** runs each ping in the background and **wait** pauses until all have finished. The addresses may appear in any order. The result lists the host (.1), Kali itself (.101) and Windows 7 (.102):

```
up: 192.168.56.1
up: 192.168.56.101
up: 192.168.56.102
```

7. netdiscover (if installed)

In one line: Finds live hosts on the local network by sending ARP requests.

Command

```
sudo netdiscover -i eth0 -r 192.168.56.0/24
```

What to look for

- **-i eth0** chooses the network card; **-r** chooses the range to scan. Use your own card name from ip a.
- This opens a live screen. Press **q** to quit.
- To print the result once and return to the prompt, add **-P**: sudo netdiscover -i eth0 -r 192.168.56.0/24 -P

Sample output (tested)

```
192.168.56.1    36:9c:38:fb:0c:74   1   42   Unknown vendor
192.168.56.102  26:b5:08:6c:1d:ab   1   42   Unknown vendor
```

```
-- Active scan completed, 2 Hosts found.
```

8. arp-scan (if installed)

In one line: Quickly lists every device on the local network with its MAC address.

Command

```
sudo arp-scan --interface=eth0 --localnet
```

What to look for

- **--localnet** scans the whole subnet of the chosen card. It needs sudo.
- It does not list Kali itself, only the other devices.

Sample output (tested)

```
Interface: eth0, type: EN10MB, MAC: 42:9d:cc:96:de:ea, IPv4: 192.168.56.101
Starting arp-scan 1.10.0 with 256 hosts
192.168.56.1     36:9c:38:fb:0c:74   (Unknown: locally administered)
192.168.56.102   26:b5:08:6c:1d:ab   (Unknown: locally administered)

2 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.913 seconds. 2 responded
```

Only run steps 7 and 8 if the tools are present in your Kali image. If they are missing, use the ping sweep in step 6.

9. nmap (quick port check)

In one line: Scans the target's ports and lists which are open (a quick, non-destructive scan).

Command

```
sudo nmap -Pn -sS -T4 192.168.56.102
```

What to look for

- **-Pn** skip the “is it alive?” check and scan anyway (needed because Windows blocks ping).
- **-sS** SYN scan: knocks on each port without finishing the connection. This needs **root**, so use **sudo**.
- **-T4** faster timing. Nmap checks the 1000 most common ports.
- **open** = a service is listening. In the sample, port 80 (web), 445 (file sharing) and 3389 (Remote Desktop) are open.

Sample output (tested)

```
Nmap scan report for 192.168.56.102
Host is up (0.0000050s latency).
Not shown: 997 closed tcp ports (reset)
PORT     STATE SERVICE
80/tcp   open  http
445/tcp  open  microsoft-ds
3389/tcp open  ms-wbt-server
MAC Address: 26:B5:08:6C:1D:AB (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 13.08 seconds
```

Tested note: without sudo, the same command stops with “You requested a scan type which requires root privileges.” If you cannot use sudo, run nmap -Pn -sT 192.168.56.102 instead (-sT makes a full connection and does not need root). The ports you see depend on which services your

Windows 7 image runs and whether its firewall is on; a firewall may show ports as filtered or hide them completely.

10. nc (test one port)

In one line: Tries to connect to a single port to see if it is open (-z = just test, -v = say what happened).

Command

```
nc -zv 192.168.56.102 80
```

What to look for

- Test other ports the same way: 3389 (Remote Desktop), 445 (file sharing).

Sample output (tested)

```
Connection to 192.168.56.102 80 port [tcp/http] succeeded!

(a closed port)
nc: connect to 192.168.56.102 port 22 (tcp) failed: Connection refused
```

succeeded = port open. **Connection refused** = port closed. A long wait with no answer usually means a firewall is blocking it.

D2. On Windows 7 (Inside the VM)

Open **Command Prompt**: Start → type cmd → Enter. These commands could not be run in the test lab, so the outputs below show the **typical** format.

1. ipconfig

In one line: Shows this Windows machine's IPv4 address, subnet mask and gateway.

Command

```
ipconfig
```

Typical output

```
Ethernet adapter Local Area Connection:
   IPv4 Address. . . . . . . . . . . : 192.168.56.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

An empty Default Gateway is normal on a Host-only network. If the IPv4 address starts with 169.254, DHCP did not work (see Troubleshooting).

2. ping 192.168.56.101

In one line: Checks that Windows 7 can reach Kali (replace the IP with your Kali address). Windows sends 4 pings by default.

Command

```
ping 192.168.56.101
```

Typical output

```
Reply from 192.168.56.101: bytes=32 time<1ms TTL=64
Reply from 192.168.56.101: bytes=32 time<1ms TTL=64
...
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Replies from both sides mean two-way connectivity is working.

3. arp -a

In one line: Shows the IP-to-MAC pairs Windows has learned (the same idea as arp -n on Kali).

Command

```
arp -a
```

Typical output

```
Interface: 192.168.56.102 --- 0xb
  Internet Address      Physical Address      Type
  192.168.56.101        08-00-27-xx-xx-xx     dynamic
```

Windows writes MAC addresses with dashes; Linux uses colons. Run it after the ping so Kali appears in the list.

Typical Troubleshooting Steps

Problem	Why it happens	What to do
ip a on Kali shows only 127.0.0.1 (no host-only IP), or hostname -I prints nothing	The adapter is not attached to the lab network, the VM was not restarted after changing settings, or the network has no DHCP.	Check VM Settings → Network → Adapter 1 is Host-only/Internal and the VM is started. Check the Host-only adapter exists (File → Host Network Manager). For an Internal Network, set a static IP (below).
ping says Network is unreachable	The machine has no IP address on that network.	Fix the IP first (see above), then ping again.
ping shows 100% packet loss; arp -n shows (incomplete) for that IP	The other machine did not answer: wrong network name, VM not running, wrong IP, or a firewall.	Verify both VMs are on the same virtual network name. Re-check the IP of the target with ipconfig.
Ping from Kali to Windows 7 fails but Windows 7 can ping Kali	Windows 7's firewall blocks incoming ping by default.	Temporarily allow ICMP in Windows Firewall (or turn the firewall off in the lab for demonstration only).
Both VMs have an IP but are in different ranges (for example 10.0.2.15)	A NAT adapter was selected by mistake. NAT isolates the VM from the host-only network.	Change the adapter to Host-only/Internal for both VMs and restart them.
netdiscover or arp-scan: command not found	The tool is not in your Kali image.	Use the ping sweep from step 6.

Setting a Static IP (Needed for Internal Network)

An Internal Network has no DHCP server, so give each VM an address by hand. Use different numbers for each machine, in the same 192.168.56.x range.

Kali (temporary until reboot):

```
sudo ip addr add 192.168.56.101/24 dev eth0
sudo ip link set eth0 up
```

Windows 7: Control Panel → Network and Sharing Center → Change adapter settings → right-click the adapter → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties → “Use the following IP address”: IP 192.168.56.102, Subnet mask 255.255.255.0 (leave gateway empty).

Command Quick Reference

Command	One-line summary
ip a	List network cards and their IP/MAC addresses.
hostname -I	Print only this machine's IP address.
ip route	Show which card reaches which network (and the gateway).
arp -n	Show learned IP-to-MAC pairs (Linux).
ping -c 4 <IP>	Send 4 test messages to check a machine replies.
for ip in 192.168.56.{1..254}; do ... done	Ping every address in the subnet and print the live ones.
sudo netdiscover -i eth0 -r 192.168.56.0/24	Find live hosts using ARP requests (live screen, q to quit).
sudo arp-scan --interface=eth0 --localnet	Quickly list all devices on the local subnet.
sudo nmap -Pn -sS -T4 <IP>	Quick scan of the target's 1000 common ports.
nc -zv <IP> <port>	Test whether one specific port is open.
ipconfig (Windows)	Show the Windows machine's IP address, mask and gateway.
ping <IP> (Windows)	Check that Windows can reach another machine (sends 4 pings).
arp -a (Windows)	Show learned IP-to-MAC pairs (Windows).

Lab Completion Checklist

- ☐ Kali network adapter checked (Host-only or Internal, same name as Windows 7).
- ☐ Windows 7 VM created, isolated and booted, snapshot taken.
- ☐ Kali IP address found with ip a / hostname -I.
- ☐ Windows 7 IP address found with ipconfig.
- ☐ Ping works from Kali to Windows 7 and from Windows 7 to Kali.
- ☐ Ping sweep (or netdiscover/arp-scan) lists both machines.
- ☐ Quick nmap or nc check shows which ports of the target are open.
- ☐ Screenshots of the above saved for submission.